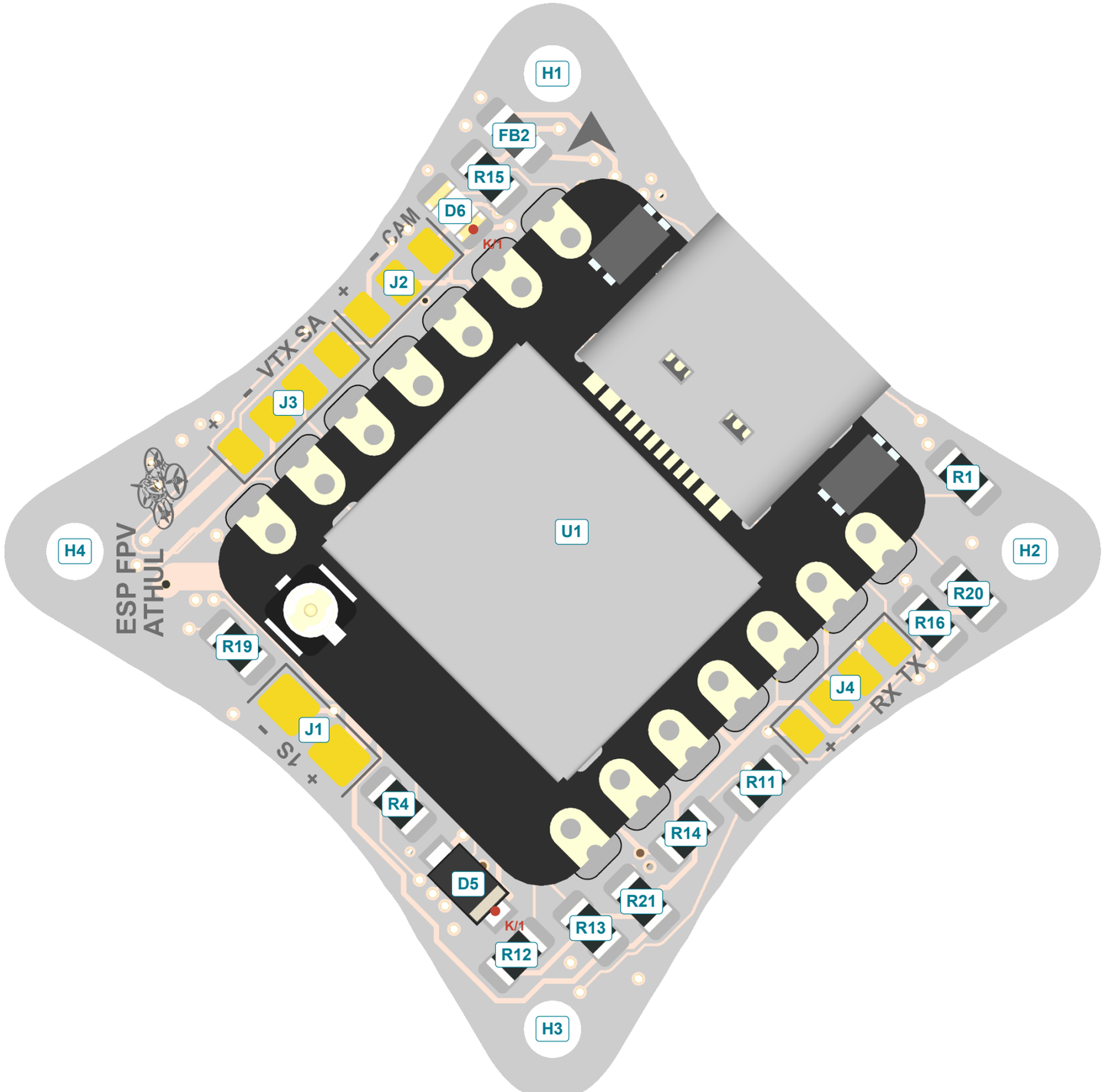
TOP ASSEMBLY / ALL COMPONENTS

Front component side facing you. Boxed references are document overlays, not extra PCB silkscreen.



23 FOOTPRINTS ON THIS SIDE

Includes four mounting holes and four external pad groups.

D5	PMEG3020ER	R4	100R
D6	BLUE 0805	R11	1k
FB2	BLM21PG601SN1D 600R	R12	10k
H1	Hole dia. 2.10 mm	R13	4.7k
H2	Hole dia. 2.10 mm	R14	10k
H3	Hole dia. 2.10 mm	R15	75R 1%
H4	Hole dia. 2.10 mm	R16	75R 1%
J1	1S BATTERY	R19	470R
J2	CAMERA	R20	100k
J3	AIR VTX	R21	4.7k 1%
J4	iBUS / ELRS CRSF receiver	U1	XIAO ESP32-S3
R1	100R		

USE THE FINAL SIDE PLACEMENT

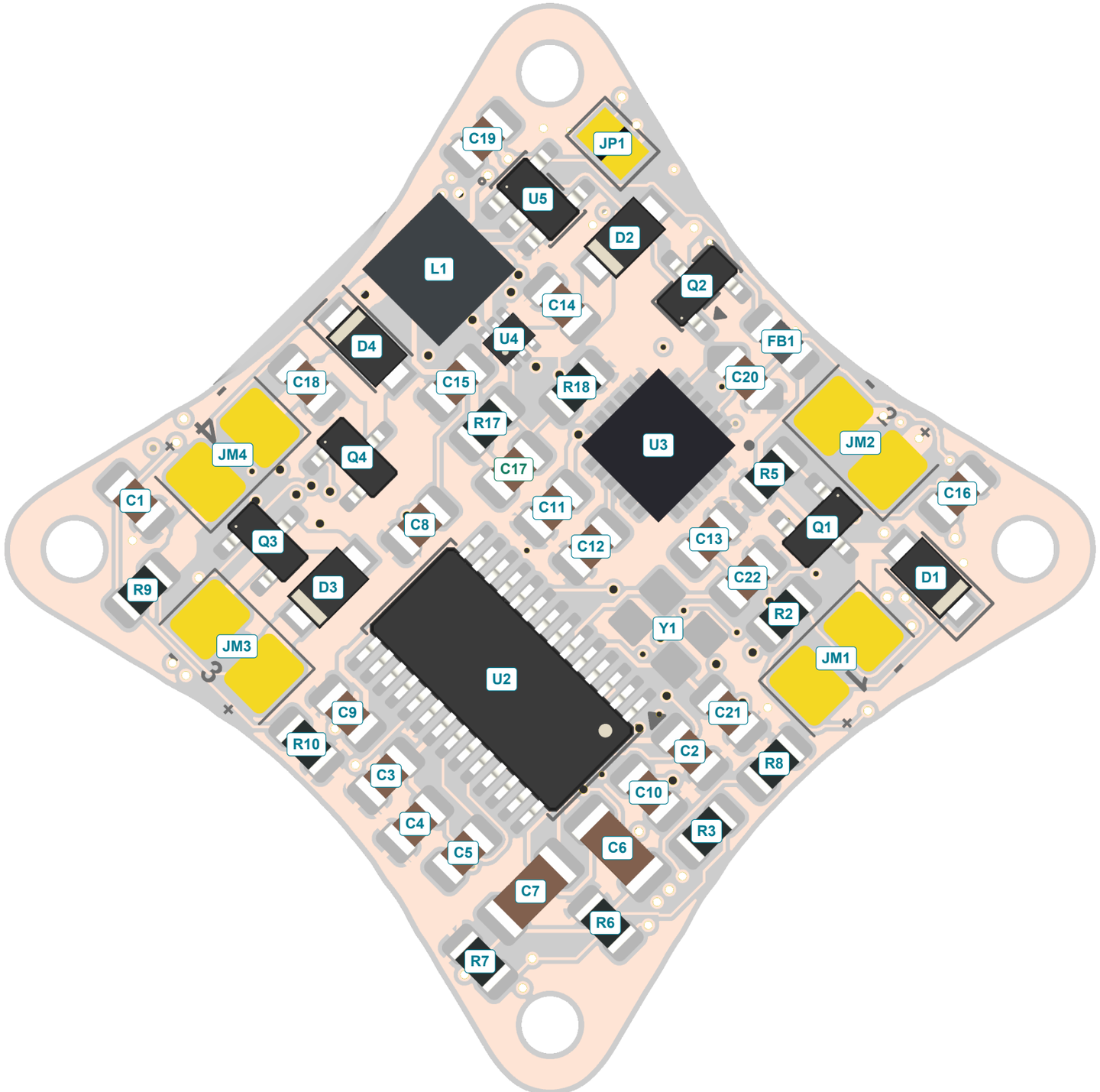
R1 and R4 are on the front; C19 is on the back.
C17 is FIT at 220 pF C0G. SDA=5, SCL=43, PWM3=39.
Front resistors, FB2 and LED D6 use 0805 packages.

SOLDER ACCESS

Fit the top-side small parts before U1. The XIAO module covers several underside contacts. Page 10 shows every U1 landing pad.

BOTTOM ASSEMBLY / ALL COMPONENTS

Back component side facing you. Left/right are reversed from the front view. Use pages 5-8 for larger closeups.



52 FOOTPRINTS ON THIS SIDE

C1	100nF	FB1	BLM21PG601SN1D 600R
C2	100nF	JM1	MOTOR 1
C3	100nF	JM2	MOTOR 2
C4	100nF	JM3	MOTOR 3
C5	100nF	JM4	MOTOR 4
C6	47uF 10V X5R	JP1	Power-isolation solder jumper
C7	47uF 10V X5R	L1	1uH XEL4030-102MEC
C8	10uF 10V X5R	Q1	AO3400A
C9	100nF	Q2	AO3400A
C10	100nF	Q3	AO3400A
C11	100nF	Q4	AO3400A
C12	100nF X7R 10V	R2	100R
C13	10nF	R3	100R
C14	10uF 10V X5R	R5	100k
C15	22uF 10V X5R	R6	100k
C16	22uF 10V X5R	R7	100k
C17	220pF C0G	R8	100k
C18	22uF 10V X5R	R9	100k 1%
C19	100nF	R10	47k 1%
C20	2.2nF X7R 50V	R17	732k 1%
C21	18pF C0G 50V	R18	100k 1%
C22	18pF C0G 50V	U2	AT7456E
D1	PMEG3020ER	U3	MPU-6050
D2	PMEG3020ER	U4	TPS61023DRLR
D3	PMEG3020ER	U5	SN74LVC1G17DBVR
D4	PMEG3020ER	Y1	YSX321SL 27MHz 12pF

ORIENTATION KEY

Red markers in the closeups identify PCB pad 1. For diodes, pad 1 is cathode K. Align the received device using its own datasheet and marking.

BATTERY, CAMERA, VTX & RECEIVER / EXACT PADS

Front side facing you. Follow each colored leader to the center of the named pad.

CAM VIDEO / J2.3
Camera video -> OSD input

CAM - / J2.2
GND

CAM + / J2.1
Assigned +5V rail

SA / J3.4
SmartAudio

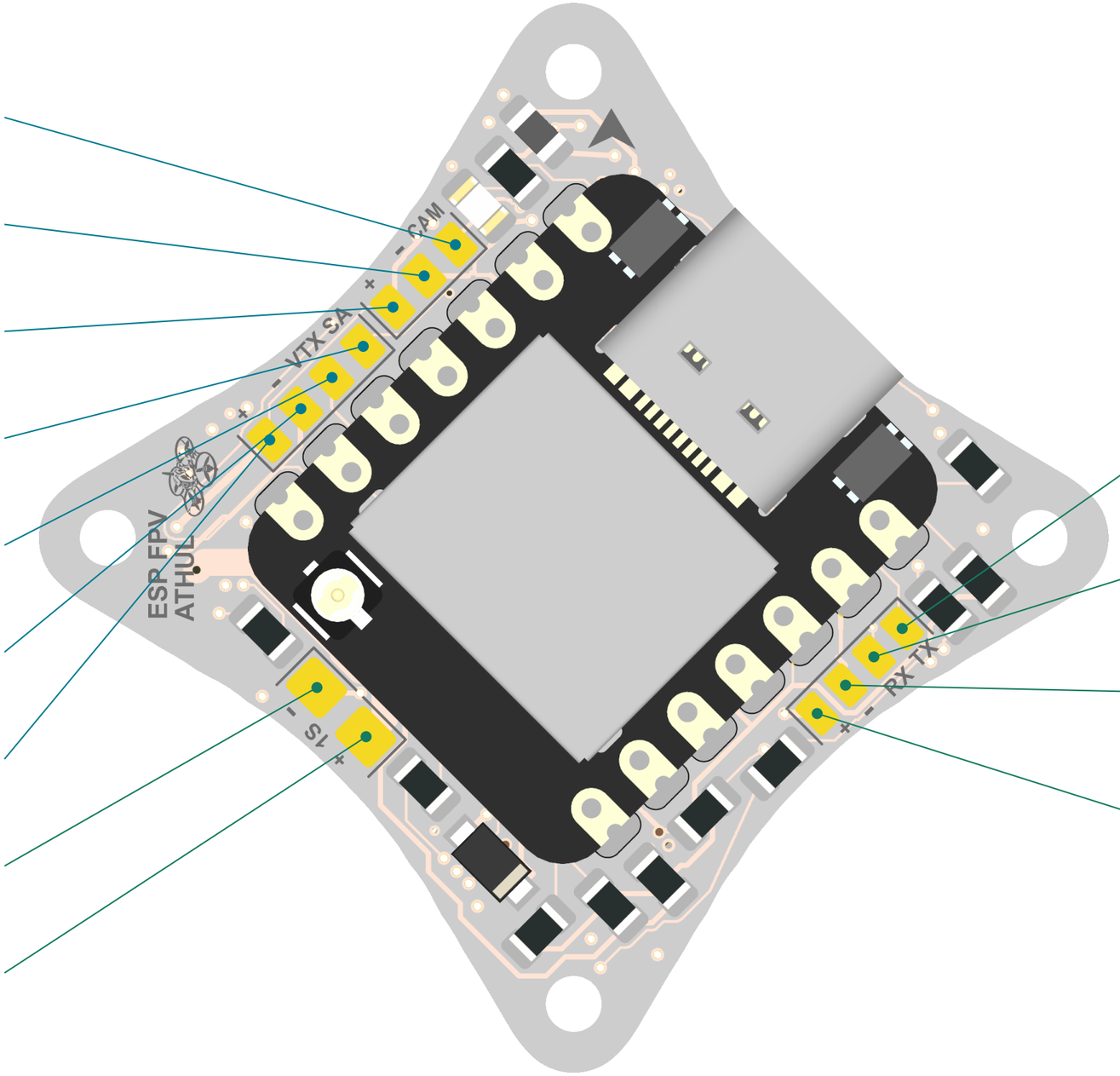
VTX VIDEO / J3.3
OSD video output

VTX - / J3.2
GND

VTX + / J3.1
Assigned +5V rail

BATTERY - / J1.2
GND

BATTERY + / J1.1
1S input / VBAT



FC TX / J4.4
Connect receiver RX here

FC RX / J4.3
Connect receiver TX here

RECEIVER - / J4.2
GND

RECEIVER + / J4.1
Assigned +5V rail

POWER AND UART

J1 takes the 1S battery. J2/J3/J4 positive pads are on the regulated +5V rail: use 5V-compatible modules. Receiver TX -> FC RX, receiver RX -> FC TX. Battery disconnected for initial USB programming.

MOTORS & POWER JUMPER / EXACT PADS

Back side facing you. Motor negative is a switched MOSFET return - it is not GND.

MOTOR 4 - / JM4.2

M4_NEG / switched return

MOTOR 4 + / JM4.1

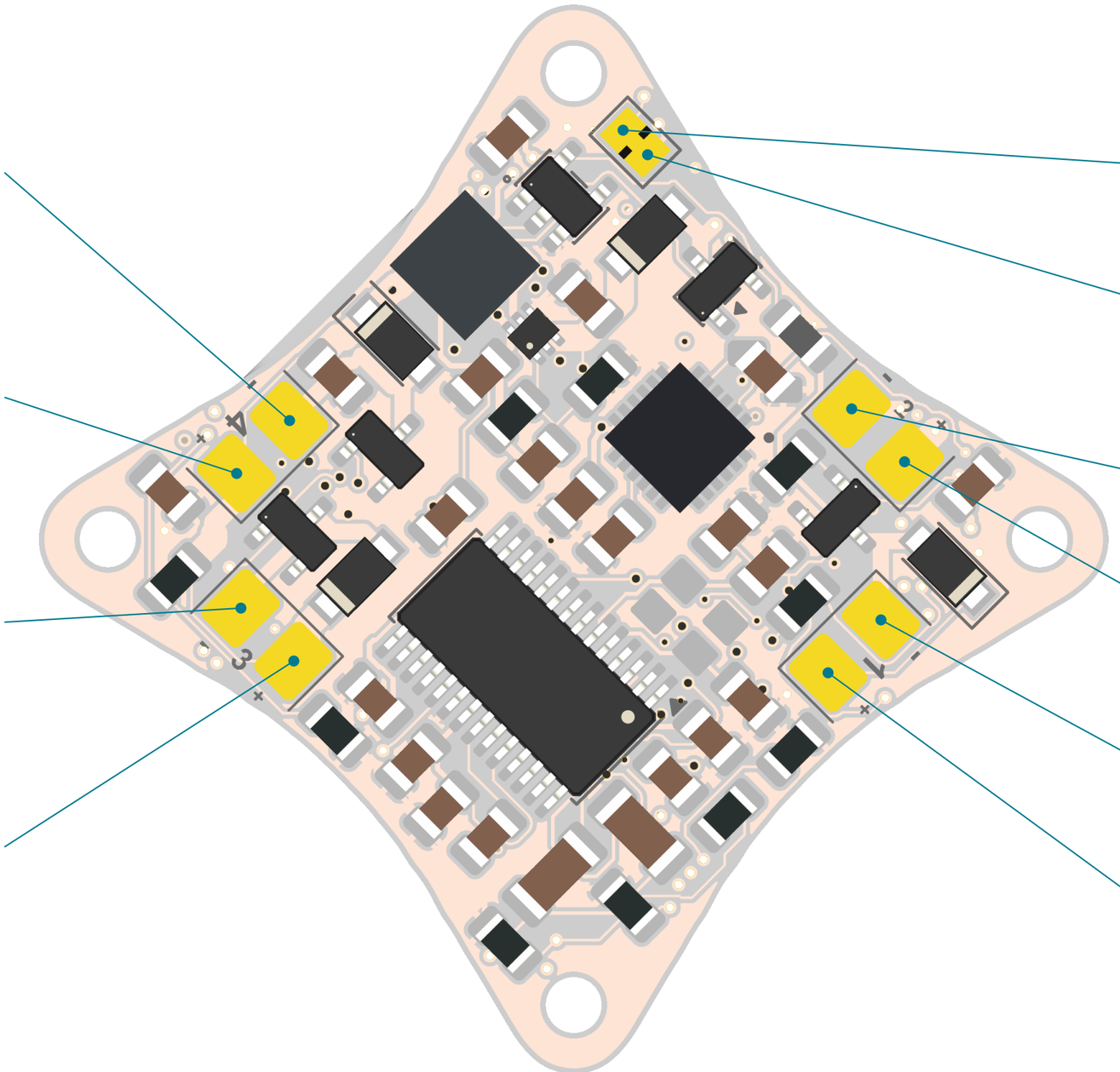
VBAT / battery positive

MOTOR 3 - / JM3.2

M3_NEG / switched return

MOTOR 3 + / JM3.1

VBAT / battery positive



POWER JUMPER / JP1.2

XIAO_5V

POWER JUMPER / JP1.1

XIAO_FEED

MOTOR 2 - / JM2.2

M2_NEG / switched return

MOTOR 2 + / JM2.1

VBAT / battery positive

MOTOR 1 - / JM1.2

M1_NEG / switched return

MOTOR 1 + / JM1.1

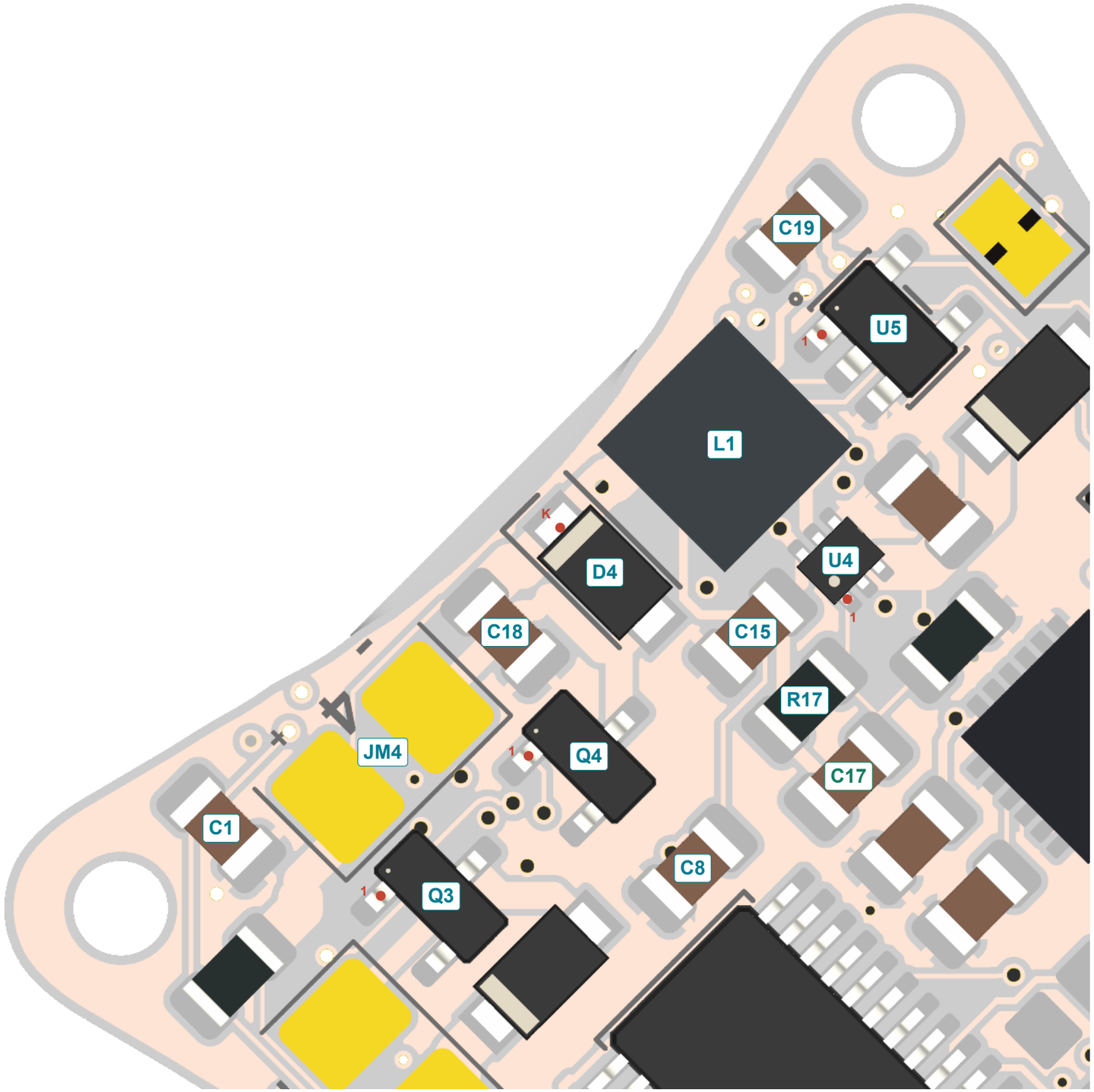
VBAT / battery positive

JP1 AND MOTOR NUMBERING

JP1 is closed for battery flight operation; open it for simultaneous USB + battery use as marked in the design. Verify its bridge state with continuity. Motor numbers are PCB labels; the firmware mixer and spin directions still need setup.

BOTTOM CLOSEUP / UPPER LEFT

Back side facing you. The crop overlaps neighboring regions so edge parts remain visible.



14 REFERENCES IN THIS REGION

C1 100nF
0805; non-polarized.

C8 10uF 10V X5R
0805; non-polarized.

C15 22uF 10V X5R
0805; non-polarized.

C17 220pF C0G
0805; FIT 220 pF C0G in L4.

C18 22uF 10V X5R
0805; non-polarized.

C19 100nF
0805; non-polarized.

D4 PMEG3020ER
Stripe/cathode K -> pad 1.

JM4 MOTOR 4
Pad 1 = VBAT+; pad 2 = switched return.

L1 1uH XEL4030-102MEC
Specified XEL4030 package; non-polarized.

Q3 AO3400A
SOT-23: 1 = gate; 2 = source; 3 = drain.

Q4 AO3400A
SOT-23: 1 = gate; 2 = source; 3 = drain.

R17 732k 1%
0805; non-polarized.

U4 TPS61023DRLR
Pin 1 = FB; hot-air/reflow recommended.

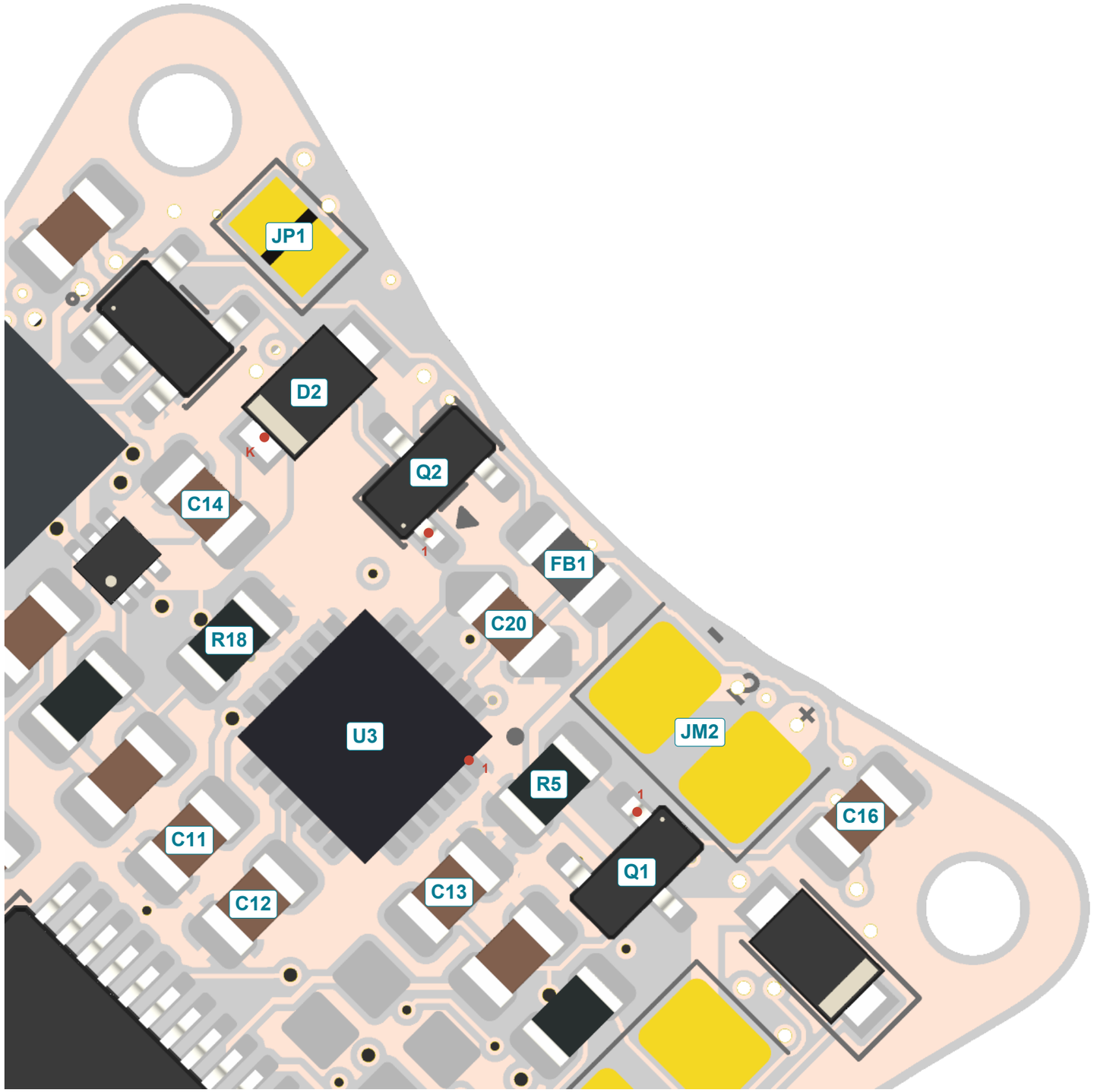
U5 SN74LVC1G17DBVR
Pin 1 = NC; align the package pin-1 mark.

BENCH KEY

C17: fit 220 pF C0G in L4. C20: 2.2 nF X7R 50 V. Use the R5 position shown here.
Red = PCB pad 1 / cathode K. Resistors and ceramic capacitors are not polarized.

BOTTOM CLOSEUP / UPPER RIGHT

Back side facing you. The crop overlaps neighboring regions so edge parts remain visible.



15 REFERENCES IN THIS REGION

C11 100nF
0805; non-polarized.

C12 100nF X7R 10V
0805; non-polarized.

C13 10nF
0805; non-polarized.

C14 10uF 10V X5R
0805; non-polarized.

C16 22uF 10V X5R
0805; non-polarized.

C20 2.2nF X7R 50V
0805; 2.2 nF X7R, 50 V; CPOUT.

D2 PMEG3020ER
Stripe/cathode K -> pad 1.

FB1 BLM21PG601SN1D 600R
0805; non-polarized.

JM2 MOTOR 2
Pad 1 = VBAT+; pad 2 = switched return.

JP1 Power-isolation solder jumper
Closed in flight; open for USB + battery.

Q1 AO3400A
SOT-23: 1 = gate; 2 = source; 3 = drain.

Q2 AO3400A
SOT-23: 1 = gate; 2 = source; 3 = drain.

R5 100k
0805; non-polarized.

R18 100k 1%
0805; non-polarized.

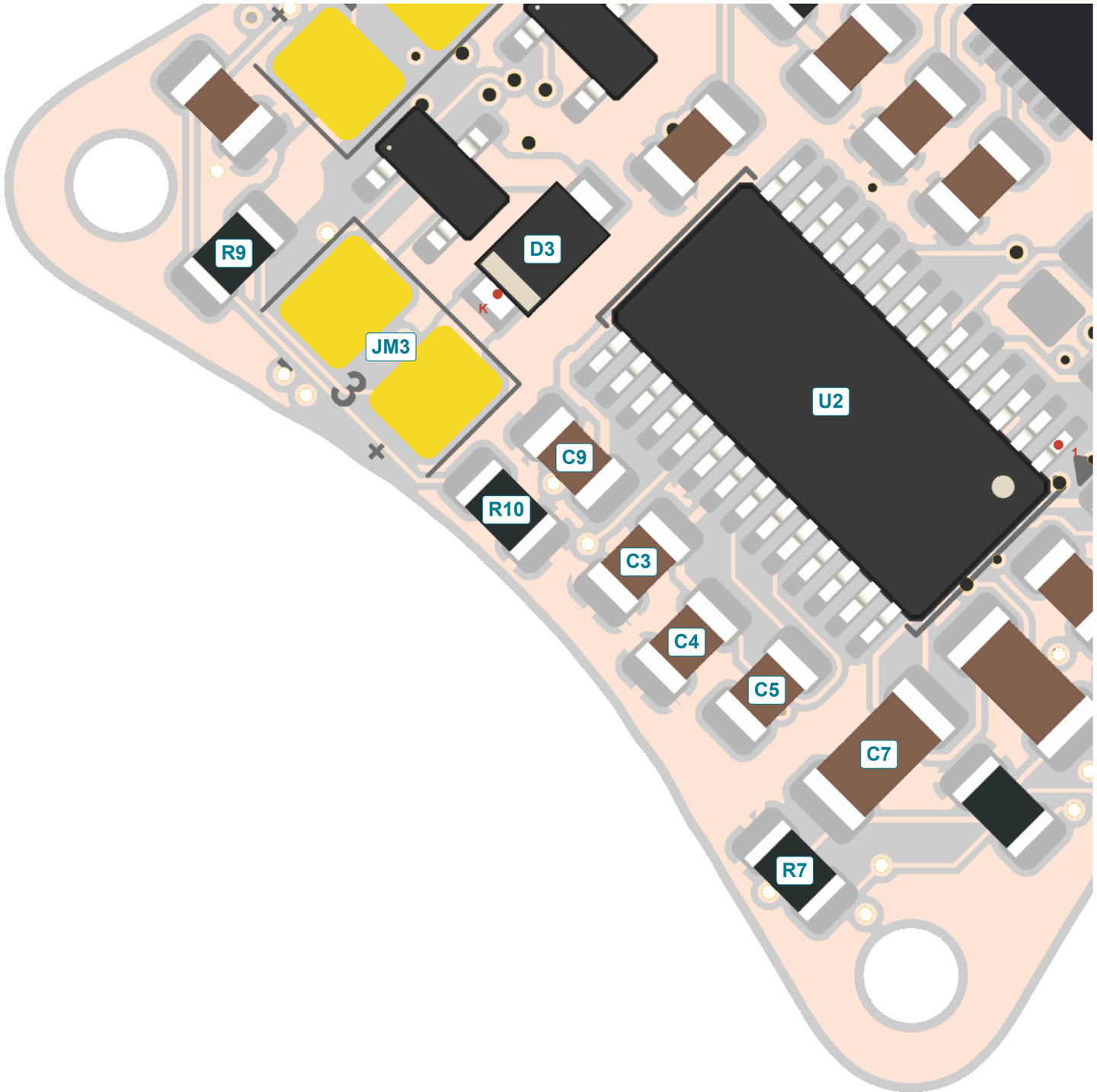
U3 MPU-6050
Bare QFN-24; NO solder under center die.

BENCH KEY

C17: fit 220 pF C0G in L4. C20: 2.2 nF X7R 50 V. Use the R5 position shown here.
Red = PCB pad 1 / cathode K. Resistors and ceramic capacitors are not polarized.

BOTTOM CLOSEUP / LOWER LEFT

Back side facing you. The crop overlaps neighboring regions so edge parts remain visible.



11 REFERENCES IN THIS REGION

C3 100nF
0805; non-polarized.

C4 100nF
0805; non-polarized.

C5 100nF
0805; non-polarized.

C7 47uF 10V X5R
1206; non-polarized.

C9 100nF
0805; non-polarized.

D3 PMEG3020ER
Stripe/cathode K -> pad 1.

JM3 MOTOR 3
Pad 1 = VBAT+; pad 2 = switched return.

R7 100k
0805; non-polarized.

R9 100k 1%
0805; non-polarized.

R10 47k 1%
0805; non-polarized.

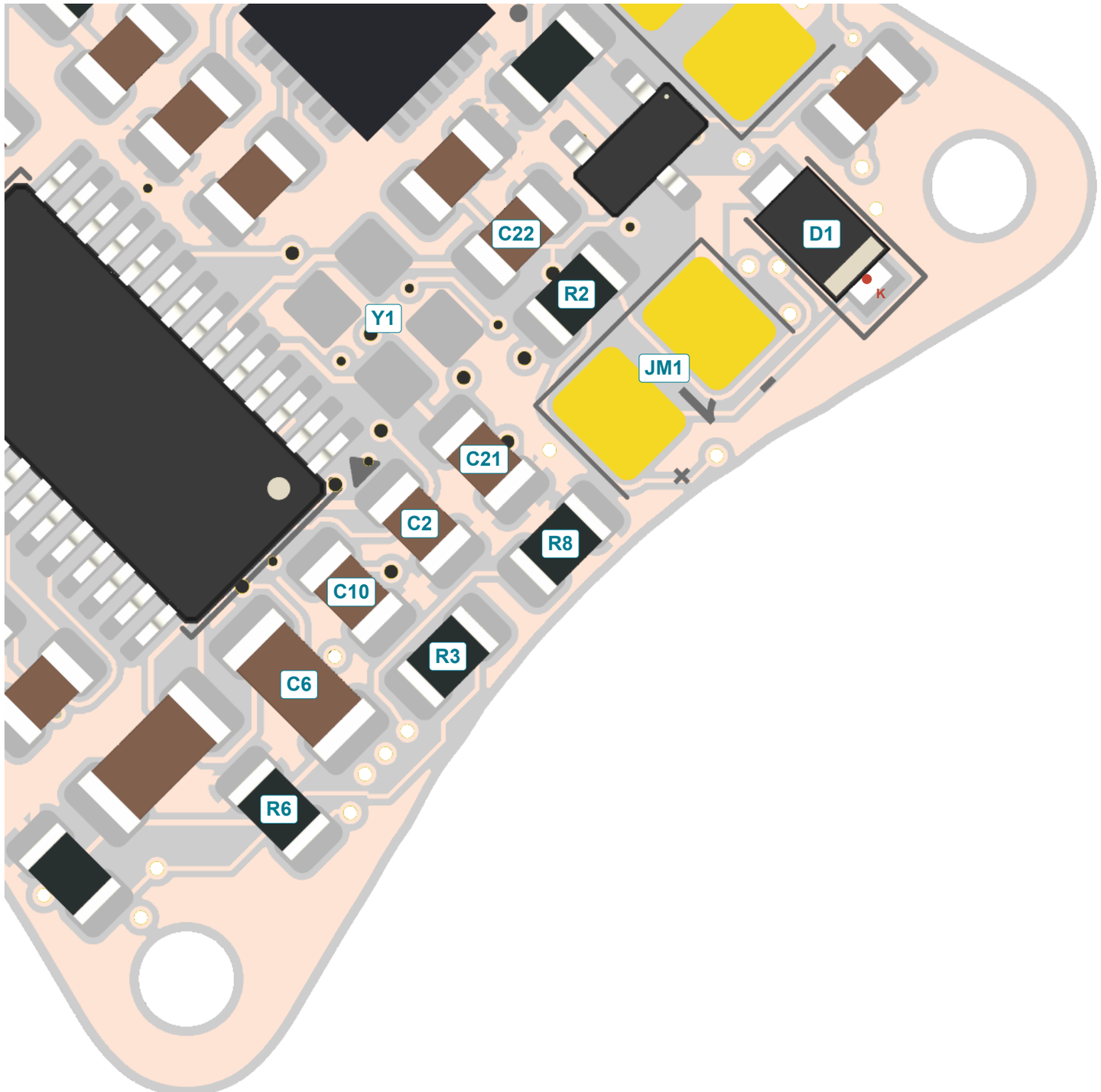
U2 AT7456E
Pin 1 marker; exposed pad 29 = GND.

BENCH KEY

C17: fit 220 pF C0G in L4. C20: 2.2 nF X7R 50 V. Use the R5 position shown here.
Red = PCB pad 1 / cathode K. Resistors and ceramic capacitors are not polarized.

BOTTOM CLOSEUP / LOWER RIGHT

Back side facing you. The crop overlaps neighboring regions so edge parts remain visible.



12 REFERENCES IN THIS REGION

C2 100nF
0805; non-polarized.

C6 47uF 10V X5R
1206; non-polarized.

C10 100nF
0805; non-polarized.

C21 18pF C0G 50V
0805; non-polarized.

C22 18pF C0G 50V
0805; non-polarized.

D1 PMEG3020ER
Stripe/cathode K -> pad 1.

JM1 MOTOR 1
Pad 1 = VBAT+; pad 2 = switched return.

R2 100R
0805; non-polarized.

R3 100R
0805; non-polarized.

R6 100k
0805; non-polarized.

R8 100k
0805; non-polarized.

Y1 YSX321SL 27MHz 12pF
3.2 x 2.5 mm passive crystal; 2/4 = GND.

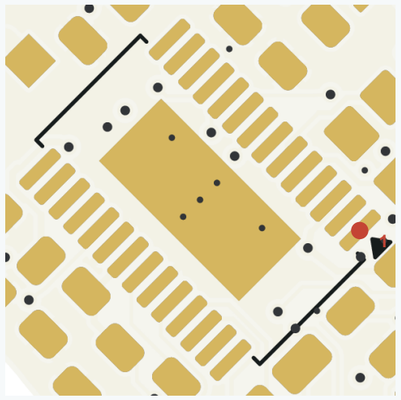
BENCH KEY

C17: fit 220 pF C0G in L4. C20: 2.2 nF X7R 50 V. Use the R5 position shown here.
Red = PCB pad 1 / cathode K. Resistors and ceramic capacitors are not polarized.

ORIENTATION / ALIGN THE REAL PACKAGE

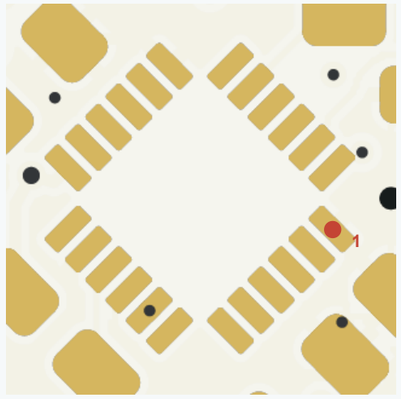
Red marker = PCB pad 1. These are component-facing views; do not mirror the received part again.

U2 / AT7456E / HTSSOP-28 + EP



Pin 1 = NC (orientation reference). Exposed pad 29 is GND and needs reflow contact.

U3 / MPU-6050 / QFN-24



Pin 1 = CLKIN, tied to GND. Do NOT add solder or paste to the exposed center die.

U4 / TPS61023 / SOT-563



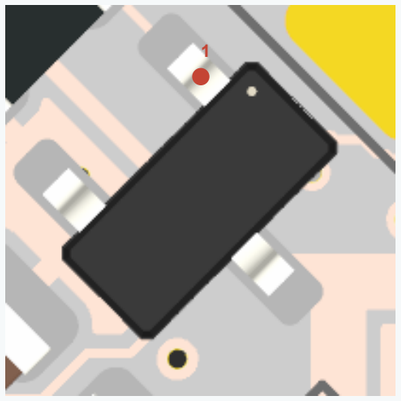
Pin 1 = FB; 4 = GND; 5 = SW; 6 = VOUT. Fine-pitch hot-air/reflow placement.

U5 / SN74LVC1G17 / SOT-23-5



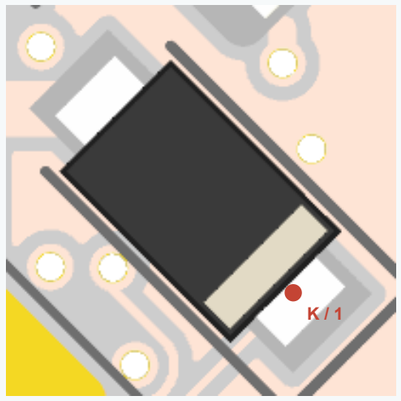
Pin 1 = NC. Align the actual package pin-1 mark; 3 = GND and 5 = VCC.

Q1 / AO3400A / Q1-Q4



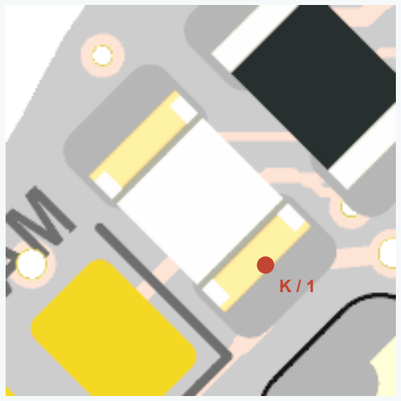
1 = gate; 2 = source; 3 = drain. Q2-Q4 have different board rotations: follow each closeup.

D1 / PMEG3020ER / D1-D5



Marking bar / cathode K goes to PCB pad 1. The opposite end is anode A, pad 2.

D6 / BLUE LED / 0805



Cathode K -> pad 1 (GND). Package marks vary; confirm with the actual LED datasheet or diode test.

Y1 / 27 MHz passive crystal

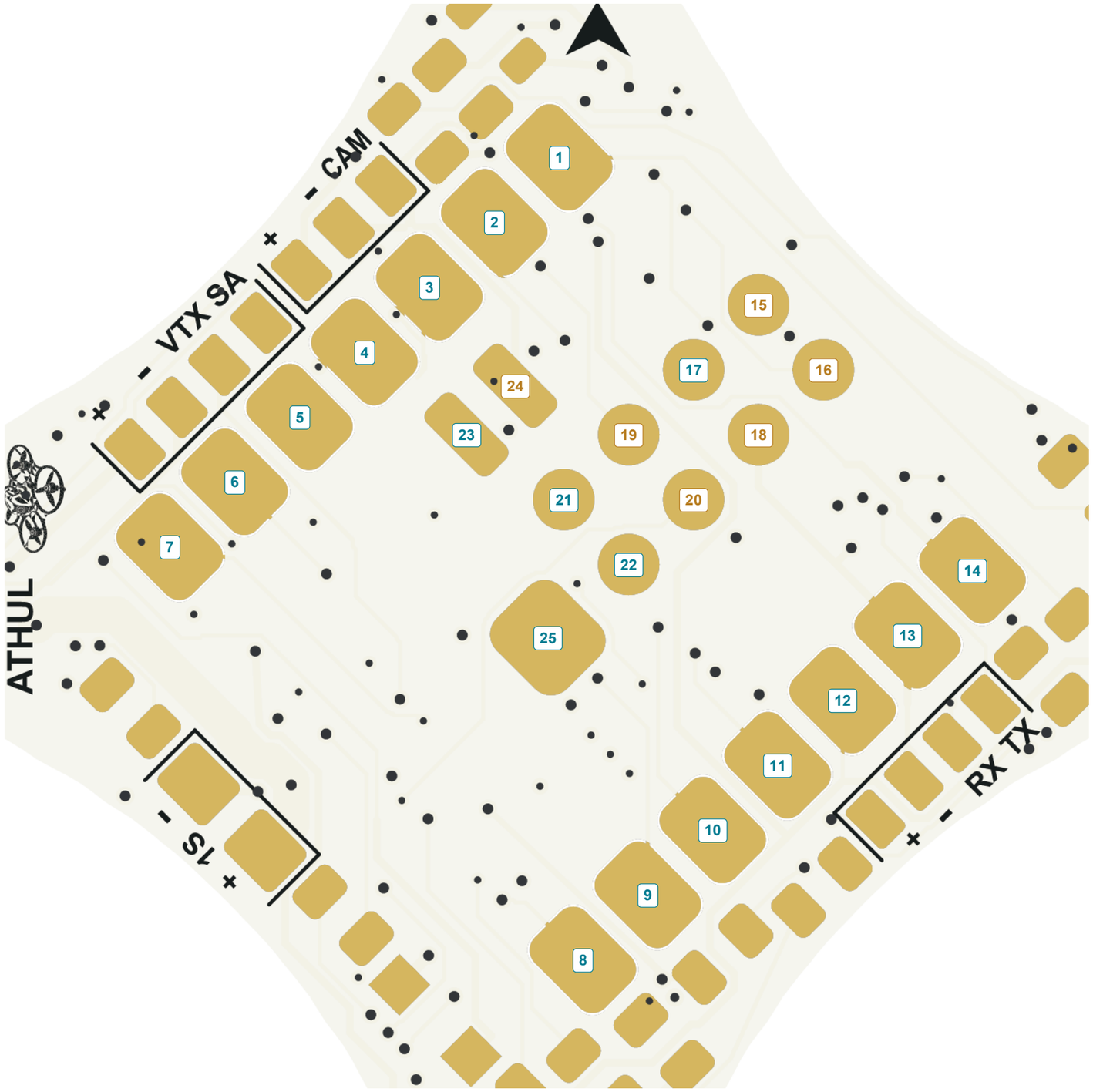


1 = OSD_XIN; 3 = OSD_XFB; 2 and 4 = GND. Use the specified four-pad passive crystal.

PMEG3020ER datasheet AO3400A datasheet TPS61023 datasheet MPU-6050 datasheet Use the received package marking and the PCB pad-1 marker.

XIAO CONTACTS / SOLDER BEFORE THE MODULE HIDES THEM

Front bare-PCB landing view. Numbers are U1 PCB pad numbers; GPIO numbers appear in the key.



U1 PAD / SIGNAL KEY

01	GPIO1 / PWM1
02	GPIO2 / PWM2
03	GPIO3 / OSD_CS
04	GPIO4 / VBAT_ADC
05	GPIO5 / IMU SDA (L4)
06	GPIO6 / PWM4
07	GPIO43 / IMU SCL (L4)
08	GPIO44 / LED
09	GPIO7 / SPI SCK
10	GPIO8 / SPI MISO
11	GPIO9 / SPI MOSI
12	3V3 output
13	GND
14	XIAO 5V / JP1
15	GPIO41 / FC TX
16	GPIO40 / FC RX
17	EN / unconnected
18	GND
19	GPIO42 / SmartAudio
20	GPIO39 / PWM3 (L4)
21	USB D- / unconnected
22	USB D+ / unconnected
23	VBAT / leave isolated
24	GND
25	Unspecified / unconnected

BEFORE MOUNTING U1

Prepare the underside GPIO and ground contacts. Use hot air/reflow where an iron cannot reach. Solder U1 last and inspect for bridges. Do not wire the battery to U1 pad 23.

Firmware: set SDA=5 and SCL=43 explicitly. Default XIAO SCL=GPIO6 is PWM4 here. Initialize PWM3 on GPIO39 low before motor power; disable/remap UART0 TX output on GPIO43.